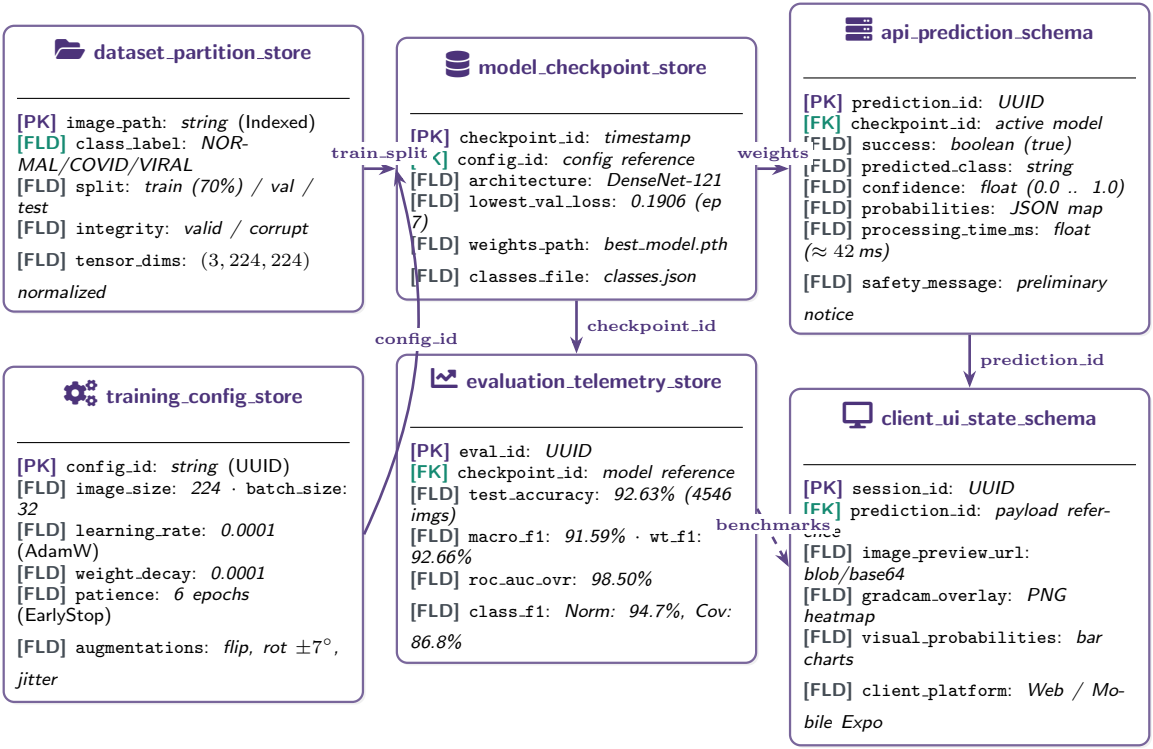

dataset_partition_store
[PK] image_path: string (Indexed)
[FLD] class_label: NORMAL/COVID/VIRAL
[FLD] split: train (70%) / val / test
[FLD] integrity: valid / corrupt
[FLD] tensor_dims: (3, 224, 224)
normalized
model_checkpoint_store
[PK] checkpoint_id: timestamp
[FK] config_id: config reference
[FLD] architecture: DenseNet-121
[FLD] lowest_val_loss: 0.1906 (ep 7)
[FLD] weights_path: best_model.pth
[FLD] classes_file: classes.json
api_prediction_schema
[PK] prediction_id: UUID
[FK] checkpoint_id: active model
[FLD] success: boolean (true)
[FLD] predicted_class: string
[FLD] confidence: float (0.0 .. 1.0)
[FLD] probabilities: JSON map
[FLD] processing_time_ms: float (≈ 42 ms)
[FLD] safety_message: preliminary notice
training_config_store
[PK] config_id: string (UUID)
[FLD] image_size: 224 · batch_size: 32
[FLD] learning_rate: 0.0001 (AdamW)
[FLD] weight_decay: 0.0001
[FLD] patience: 6 epochs (EarlyStop)
[FLD] augmentations: flip, rot ±7°, jitter
evaluation_telemetry_store
[PK] eval_id: UUID
[FK] checkpoint_id: model reference
[FLD] test_accuracy: 92.63% (4546 imgs)
[FLD] macro_f1: 91.59% · wt_f1: 92.66%
[FLD] roc_auc_ovr: 98.50%
[FLD] class_f1: Norm: 94.7%, Cov: 86.8%
client_ui_state_schema
[PK] session_id: UUID
[FK] prediction_id: payload reference
[FLD] image_preview_url: blob/base64
[FLD] gradcam_overlay: PNG heatmap
[FLD] visual_probabilities: bar charts
[FLD] client_platform: Web / Mobile Expo
train_split
weights
config_id
checkpoint_id
prediction_id
benchmarks